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REDUCED-ORDER LONGITUDINAL CONTROLLABILITY USING LINE-OF-SIGHT LANDING ERROR IN LOW-LIFT ATMOSPHERIC RE-ENTRY

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This paper examines longitudinal landing controllability for low-lift atmospheric re-entry vehicles using a line-of-sight (LOS) surface-error framework and bounded pitch authority. An onboard predictor estimates terminal landing location during flight, while pitch-induced aerodynamic effects are represented through a reduced-order formulation that captures longitudinal sensitivity without bank modulation or full attitude-state integration. Numerical simulations of a representative high-mass configuration demonstrate that both undershoot and overshoot dispersions of several hundred kilometers can be reshaped into a 50 km landing uncertainty region while remaining within angle-of-attack envelopes and load levels observed in simulation. The results highlight the extent of achievable downrange modulation using pitch-only control and provide a controllability-focused baseline for early-stage re-entry vehicle design.

INTRODUCTION

Precision atmospheric re-entry and landing of large, low-lift vehicles remains a challenging problem for heavy payload return and reusable transportation concepts. Many emerging architectures operate with a limited lift-to-drag ratio, constrained control authority, and strict actuator saturation limits. Under such conditions, classical three-dimensional entry guidance strategies based on bank-angle modulation or aggressive trajectory shaping may be impractical during early design phases where simplicity and robustness are prioritized.

Traditional entry guidance methods, including predictor–corrector schemes developed for Apollo-era vehicles,[1,2] regulate downrange primarily through lift-vector rotation and bank reversals. Modern approaches extend these concepts using optimization, model predictive control, or learning-based techniques, often requiring accurate onboard models and significant computational resources. While effective for high-lift vehicles, these approaches are not always aligned with low-lift configurations that possess limited lateral authority and restricted actuation.

This work investigates a reduced-order longitudinal guidance concept based on pitch-induced aerodynamic effects. A landing-site–referenced line-of-sight (LOS) surface error is computed during flight together with a forward prediction of the terminal landing location. In the present study, the predictor serves primarily as a trajectory awareness metric rather than a direct closed-loop driver. Control authority is represented through angle-of-attack envelope–based pitch modulation, where commanded pitch moments are mapped into an equivalent longitudinal perturbation used to examine downrange sensitivity. Figure 1 provides a conceptual illustration of the pitch-induced longitudinal range modulation mechanism examined in this study.

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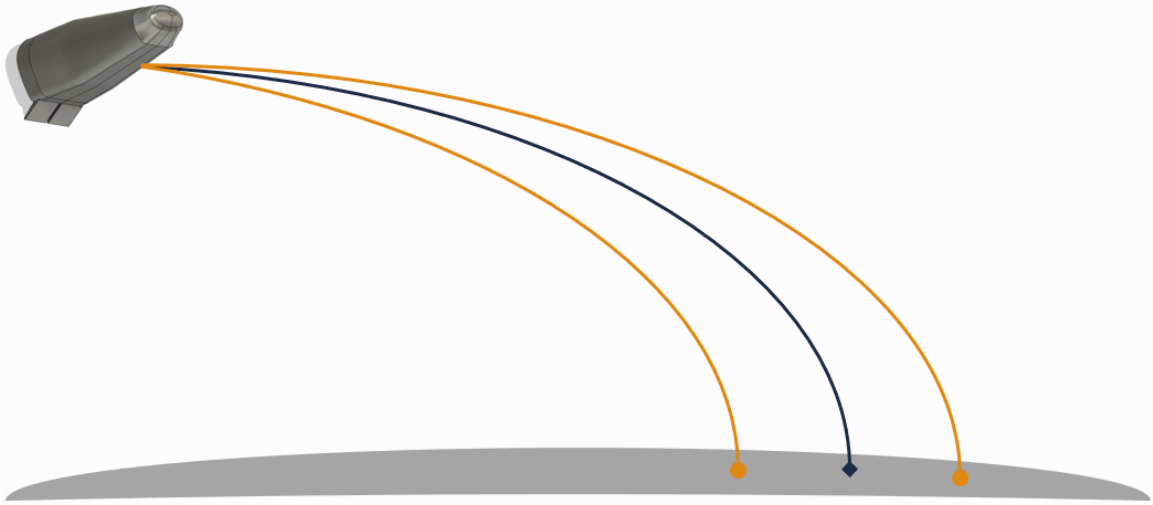


Figure 1: Conceptual illustration of pitch-induced longitudinal range modulation for a low-lift atmospheric re-entry vehicle. A nominal trajectory (center) can be reshaped into undershoot or overshoot solutions through bounded pitch authority without bank-angle modulation. The schematic highlights how small longitudinal perturbations modify integrated downrange distance rather than lateral direction.

Rather than integrating full pitch-attitude dynamics, the formulation adopts a reduced-order representation that captures the dominant coupling between aerodynamic pitch moment and longitudinal motion. This abstraction enables a computationally light framework suitable for early-stage controllability assessment while remaining consistent with actuator limits and aerodynamic constraints.

The contributions of this paper are summarized as follows:

1. Formulation of a LOS-based surface landing error metric with onboard terminal prediction for longitudinal re-entry analysis.
2. Introduction of a pitch-moment reduced-order formulation that represents bounded control authority without requiring full attitude-state integration.
3. Numerical demonstration of downrange sensitivity to pitch-only modulation for a representative low-lift re-entry configuration.

The scope of this work is limited to longitudinal motion and downrange behavior along the approximate LOS to the landing site. Crossrange control, bank modulation, and atmospheric disturbances are not explicitly modeled. The objective is to establish a minimal, controllability-focused baseline aligned with early vehicle design studies. This work is intended as a controllability-focused engineering study rather than a complete flight guidance architecture.

REDUCED-ORDER VEHICLE MODEL AND LOS ERROR FRAMEWORK

The reduced-order formulation intentionally avoids modeling pitch attitude states; instead, commanded pitch moment is mapped to an equivalent longitudinal perturbation used only to examine

the sensitivity of downrange motion. We consider a representative high-mass, low-lift atmospheric re-entry vehicle operating near zero bank with limited lateral authority. The analysis focuses on longitudinal motion and downrange behavior along the approximate line of sight (LOS) to a designated landing site. Rather than modeling full six-degree-of-freedom attitude dynamics, a reduced-order formulation is adopted in which pitch control authority is represented through a bounded effective longitudinal perturbation acting on flight-path angle dynamics.

Translational Dynamics

Let r denote radial distance from the Earth's center, ϕ latitude, λ longitude, v inertial speed, and γ flight-path angle. The reduced point-mass equations of motion in spherical coordinates are[3,4]

$$\dot{v} = -\frac{D}{m} - g(r)\sin\gamma, \tag{1}$$

$$\dot{\gamma} = \frac{L}{mv} - \left(\frac{g(r)}{v} - \frac{v}{r} - \frac{T_r \sin\alpha}{mv\cos\gamma}\right)\cos\gamma, \tag{2}$$

$$\dot{r} = v\sin\gamma, \tag{3}$$

$$\dot{\lambda} = \frac{v\cos\gamma\cos\chi}{r\cos\phi}, \tag{4}$$

$$\dot{\phi} = \frac{v\cos\gamma\sin\chi}{r}, \tag{5}$$

where L and D denote aerodynamic lift and drag, T_r is the thrust in the radial direction.[5] Rather than appearing as an additive term, pitch authority enters the flight-path-angle dynamics through a modified longitudinal coupling inside the curvature–gravity bracket, consistent with the reduced-order implementation used in simulation. Near-zero bank is assumed so that lateral coupling remains secondary.

Aerodynamic forces are modeled as

$$L = \frac{1}{2}\rho v^2 S C_L(\alpha), \tag{6}$$

$$D = \frac{1}{2}\rho v^2 S C_D(\alpha), \tag{7}$$

where ρ is atmospheric density and α is the instantaneous angle of attack. In the present formulation, α is treated as a reduced-order kinematic proxy derived from translational motion rather than from explicit attitude dynamics; control action acts only when the proxy angle approaches prescribed bounds.

Pitch Authority Representation

Rather than integrating explicit pitch-attitude states, control authority is represented through a bounded pitching moment formulation. Aerodynamic pitching moment is modeled as

$$M_{\text{aero}} = \frac{1}{2}\rho v^2 S\bar{c}\, C_m(\alpha). \tag{8}$$

Control-generated moments M_{ctrl} are limited by angle-of-attack constraints and mapped into a longitudinal perturbation through a fixed moment-arm approximation,

$$T_r = K_r \frac{M_{\text{ctrl}}}{\ell}, \tag{9}$$

where ℓ is an effective moment arm and K_r is a constant guidance gain mapping commanded pitch moment into an equivalent longitudinal perturbation. In the present reduced-order model, this mapping is linear and is not derived from explicit attitude dynamics.

Landing Coordinate Error

Let (ϕ^*, λ^*) denote the target landing site. The great-circle angular separation between the vehicle subpoint and the target is σ, and the surface landing error is

$$\rho = R_e \sigma. \tag{10}$$

In addition to instantaneous LOS error, a forward propagation of the current state provides a predicted terminal landing error $\hat{\rho}$. In this study, $\hat{\rho}$ serves as a trajectory awareness metric used to interpret guidance behavior rather than as a closed-loop control signal.

Operational Constraints

Simulation cases are evaluated against prescribed angle-of-attack envelopes,

$$\alpha_{\min} \leq \alpha \leq \alpha_{\max}, \tag{11}$$

together with limits on dynamic pressure and load factor,

$$q = \frac{1}{2}\rho v^2 \leq q_{\max}, \tag{12}$$

$$n = \frac{\|\mathbf{a}\|}{g_0} \leq n_{\max}. \tag{13}$$

Within the observed envelopes, pitch-induced perturbations provide a monotonic influence on longitudinal motion, enabling evaluation of downrange controllability for low-lift configurations.

The resulting formulation defines a minimal longitudinal framework for analyzing LOS landing error, terminal prediction, and bounded pitch authority without reliance on bank modulation, full trajectory optimization, or explicit attitude dynamics.

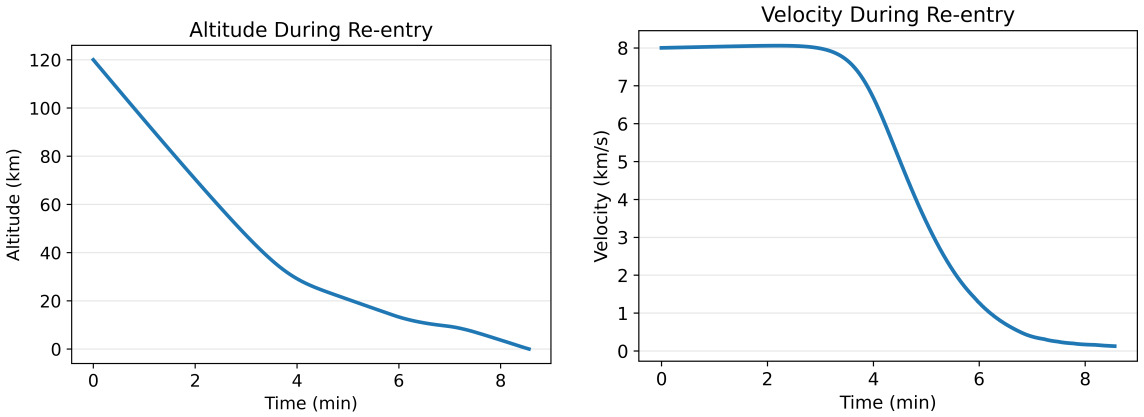
NUMERICAL EVALUATION OF LONGITUDINAL CONTROLLABILITY

Nonlinear trajectory analysis is used to examine achievable downrange modulation using bounded pitch authority under representative low-lift re-entry conditions. The objective is not trajectory optimization, but assessment of controllability trends under realistic aerodynamic limits and path constraints.

Vehicle and Entry Conditions

The vehicle corresponds to a high-mass re-entry configuration with a lift-to-drag ratio $L/D \in [0.2, 0.5]$. Aerodynamic coefficients $C_L(\alpha)$, $C_D(\alpha)$, and $C_m(\alpha)$ are provided as tabulated data and interpolated during integration. Atmospheric density follows a standard Earth reference profile, and gravity is modeled as $g(r) = \mu_\oplus / r^2$.

At each integration step, the state is forward propagated to estimate the terminal landing location. The resulting predicted landing error serves as a trajectory awareness metric, while bounded pitch authority is represented through angle-of-attack envelopes and gain-limited control moments within the reduced-order model. Two dispersion classes are considered: undershoot trajectories landing short of the target and overshoot trajectories exceeding the target downrange. The objective is containment within a 50 km landing uncertainty region while remaining within aerodynamic and load levels observed in simulation.



(a) Altitude history during terminal re-entry. Rapid altitude loss coincides with increasing atmospheric density and growing aerodynamic control authority, marking the regime in which pitch-only modulation becomes effective for longitudinal range correction.

(b) Velocity history during terminal re-entry. Aerodynamic deceleration increases sharply as the vehicle enters denser atmospheric layers, enabling effective modulation of downrange motion through angle-of-attack control.

Figure 2: Altitude and velocity time histories for a representative undershoot correction case. The transition into a denser atmosphere provides the aerodynamic authority required for effective LOS-based pitch-only range modulation.

Atmospheric Transition and Guidance Response

Figure 2 shows representative altitude and velocity histories. As dynamic pressure increases during atmospheric transition, aerodynamic authority grows rapidly, and pitch-induced longitudinal effects become significant. Figure 3 compares the instantaneous LOS error with the predicted terminal landing error. Transient variations in predicted terminal error arise from trajectory reshaping, while LOS error evolves smoothly toward the target region.

Angle-of-attack and deceleration histories in Fig. 4 demonstrate that meaningful downrange modulation occurs without sustained constraint saturation. Increasing utilization of available aerodynamic authority is observed during the terminal phase.

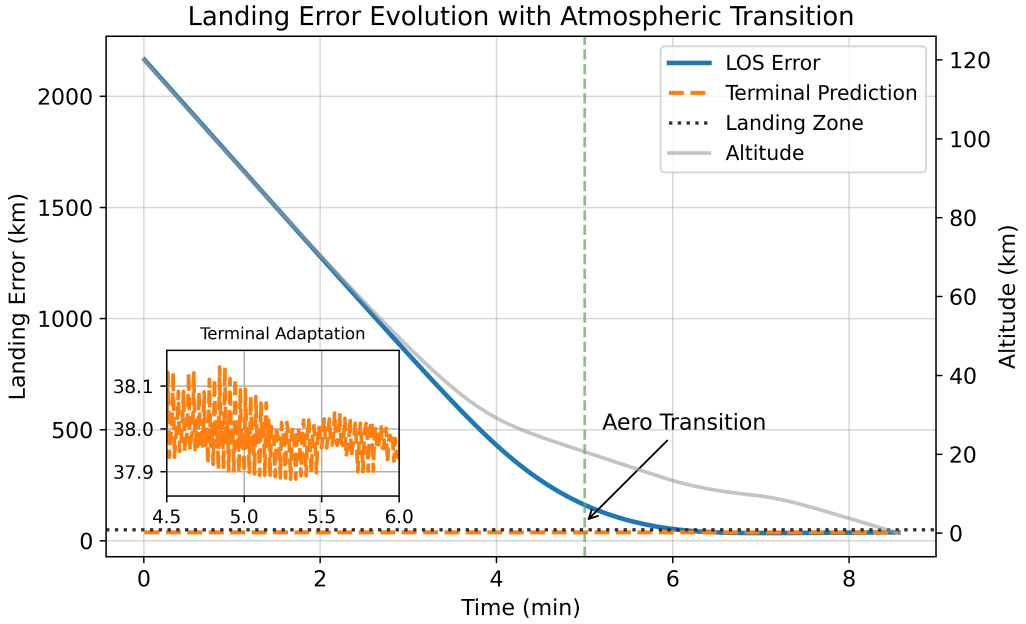
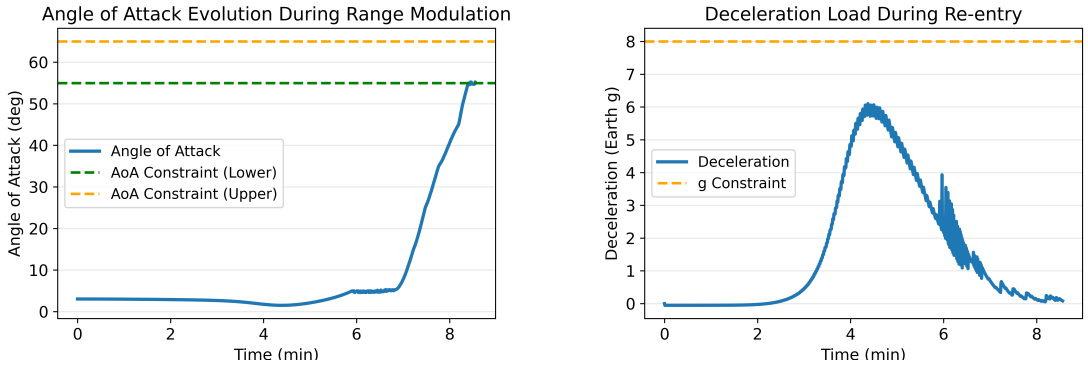


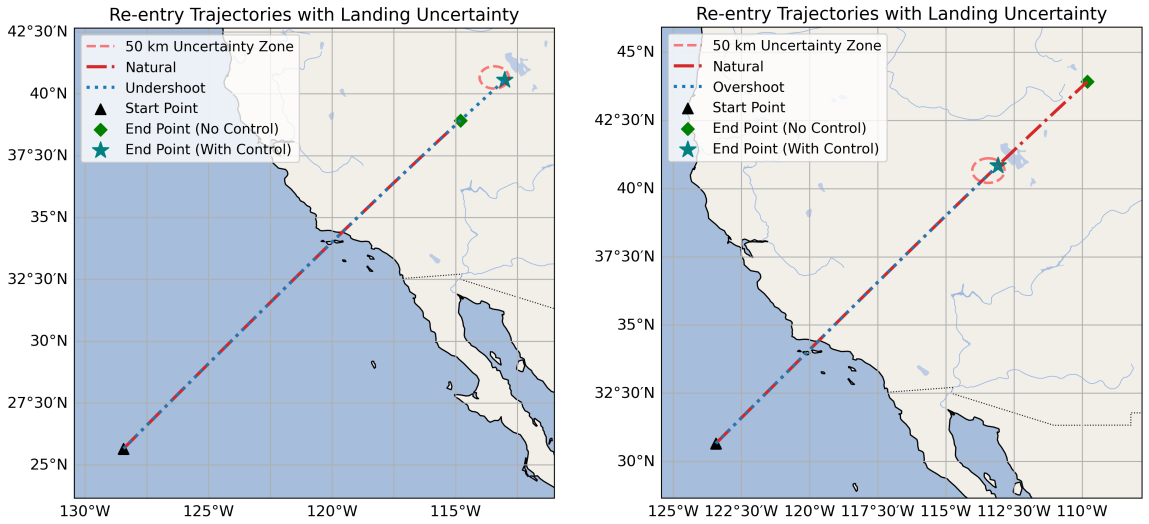
Figure 3: Instantaneous LOS landing error and predicted terminal landing error during terminal re-entry. Transient variations in predicted terminal error occur during atmospheric transition due to rapid changes in aerodynamic authority and trajectory reshaping. The instantaneous LOS error exhibits a predominantly decreasing trend, indicating stable longitudinal reshaping behavior.



(a) Angle-of-attack evolution during range modulation. The trajectory approaches prescribed limits during the terminal phase, indicating increasing utilization of available aerodynamic authority without sustained saturation.

(b) Resulting deceleration history during longitudinal reshaping. Load levels remain within allowable limits throughout the maneuver, confirming constraint-compliant pitch-only control during terminal entry.

Figure 4: Angle-of-attack evolution and corresponding deceleration for a representative undershoot correction case. Effective range modulation occurs during atmospheric transition while remaining within aerodynamic and load levels observed in simulation.



(a) Undershoot correction using LOS-based pitch-only longitudinal modulation. The nominal uncontrolled trajectory falls short of the target, while guidance-enabled angle-of-attack modulation lengthens the trajectory by approximately 175 km, achieving landing within the prescribed uncertainty zone.

(b) Overshoot correction using LOS-based pitch-only longitudinal modulation. The uncontrolled trajectory overshoots the target, while increased angle-of-attack modulation shortens the trajectory by approximately 420 km, achieving terminal landing accuracy within tens of kilometers.

Figure 5: Ground-track comparison for undershoot and overshoot correction scenarios. Starting from a common entry region and descending along the LOS direction, pitch-only modulation enables both range extension and reduction without bank control.

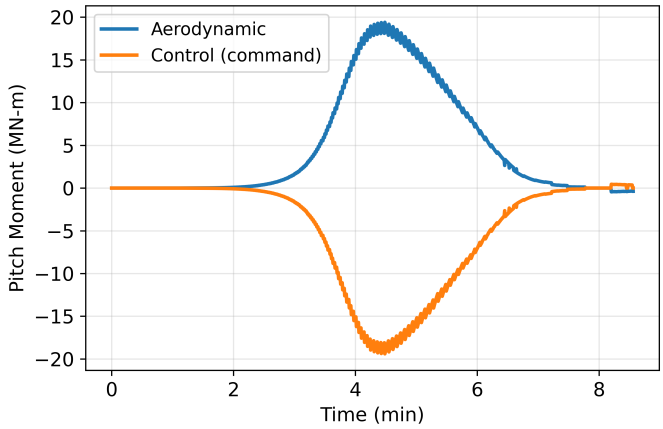


Figure 6: Pitching moment and commanded control torque history during terminal re-entry. Commanded control torque remains smooth and within the representative ranges used in the reduced-order study while producing the observed angle-of-attack modulation during terminal re-entry.

Downrange Modulation Along LOS

Ground tracks for representative undershoot and overshoot dispersions are shown in Fig. 5. Sustained pitch modulation reshapes the trajectory along the LOS direction, enabling both range extension and range reduction without bank control. In the undershoot case, longitudinal modulation lengthens the trajectory by approximately 175 km, while in the overshoot case, increased aerodynamic deceleration shortens the trajectory by roughly 420 km. In both scenarios, final landing locations fall within the prescribed uncertainty region.

Pitch moment histories (Fig. 6) indicate bounded control effort throughout terminal descent. Control actions remain smooth and consistent with the reduced-order authority levels represented in the model, confirming that the reduced-order pitch authority representation captures the dominant longitudinal coupling.

Gain Sensitivity

A parametric sweep of guidance gain illustrates qualitative controllability trends (Fig. 7). Increasing gain amplifies the effective longitudinal perturbation induced by bounded pitch authority, improving landing accuracy while increasing proximity to angle-of-attack limits. The results illustrate a trade between terminal precision and actuator margin rather than an optimal gain schedule.

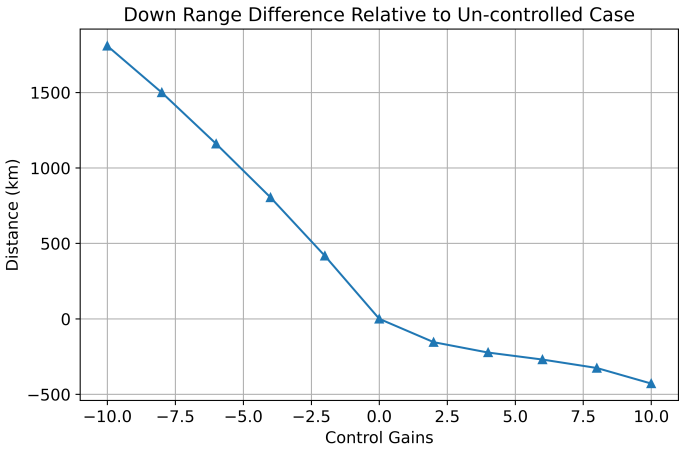


Figure 7: Sensitivity of terminal landing error and peak angle-of-attack usage to guidance gain selection. Increasing gain increases the effective longitudinal perturbation induced by bounded pitch authority, but increases control effort and proximity to path constraints, illustrating the trade between precision and actuator margin. Downrange correction is achieved as a function of guidance gain relative to the uncontrolled trajectory.

DISCUSSION AND CONCLUSIONS

This study examined longitudinal landing controllability for low-lift atmospheric re-entry vehicles using a reduced-order framework based on LOS surface error and bounded pitch authority. Numerical results indicate that substantial downrange modulation can be achieved through pitch-induced aerodynamic effects alone, enabling correction of large undershoot and overshoot dispersions without bank modulation.

The observed downrange reshaping can be interpreted physically through the sensitivity of flight-path angle dynamics to longitudinal aerodynamic coupling during atmospheric transition. As dynamic pressure increases, small pitch-induced changes to the effective longitudinal perturbation alter the balance between gravitational curvature and lift-induced trajectory bending, modifying the vehicle's time spent at high-altitude energy states. This produces a cumulative shift in integrated downrange distance rather than an instantaneous redirection of motion. In undershoot cases, reduced effective descent rate extends time-of-flight and increases ground-track length, whereas in overshoot cases increased aerodynamic loading accelerates energy dissipation and shortens the trajectory. The LOS-based predictor does not directly command control but reveals this evolving sensitivity, highlighting the role of atmospheric transition as the primary leverage point for longitudinal controllability in low-lift configurations.

Effective controllability emerges primarily during atmospheric transition, when increasing dynamic pressure amplifies the influence of pitch authority on flight-path angle evolution. Transient fluctuations in predicted terminal error arise from rapid trajectory reshaping rather than loss of stability, while instantaneous LOS error decreases consistently toward the target region.

The formulation intentionally avoids full attitude-state integration and instead represents pitch authority through an effective longitudinal perturbation. This abstraction enables controllability assessment with minimal modeling complexity, making the approach suitable for early-stage vehicle studies with constrained actuation.

The present work is limited to near-zero bank operation and does not explicitly model atmospheric disturbances, sensing errors, or lateral guidance. Rather than presenting a complete flight-ready guidance architecture, the results establish a baseline understanding of achievable longitudinal control authority for heavy, low-lift configurations.

Future work will extend this framework to include disturbance modeling, adaptive gain selection, lateral guidance augmentation, and higher-fidelity aerodynamic representations.

APPENDIX: QUALITATIVE BOUNDEDNESS INTERPRETATION

This appendix provides a qualitative interpretation of the bounded landing-error behavior observed in simulation. The objective is not to establish formal stability of the nonlinear re-entry dynamics, but to explain the monotonic trends and ultimate containment of line-of-sight (LOS) landing error under bounded pitch authority.

Landing Error Evolution

Let $\rho(t)$ denote the instantaneous LOS surface landing error. In the present formulation, ρ does not follow a prescribed differential equation and is not directly regulated by a closed-loop control law. Instead, pitch authority is applied through bounded angle-of-attack modulation, and longitudinal trajectory changes emerge from the coupled aerodynamics.

The landing-error rate may be expressed abstractly as

$$\dot{\rho} = f(\rho, x(t)), \tag{14}$$

where $x(t)$ represents the translational state variables. During atmospheric transition, increasing dynamic pressure amplifies the sensitivity of downrange motion to pitch authority, producing the monotonic trends observed in simulation.

Boundedness Sketch

Consider the nonnegative function

$$V(\rho) = \frac{1}{2}\rho^2. \tag{15}$$

Its time derivative along trajectories is

$$\dot{V} = \rho\dot{\rho}. \tag{16}$$

Simulation results indicate that outside a prescribed landing corridor, pitch modulation produces an empirically observed tendency for $|\rho|$ to decrease as aerodynamic authority increases. This behavior may be interpreted as

$$\rho\dot{\rho} \leq 0 \quad \text{for } |\rho| > \rho_{\text{tol}}, \tag{17}$$

where ρ_{tol} denotes the landing uncertainty region. Within this region, constraint saturation limits further correction, and ρ remains bounded. This may be viewed as a Lyapunov-like boundedness intuition,[6] with conceptual parallels to relaxed derivative viewpoints discussed by Mukherjee and Chen,[7] though no formal stability proof is claimed.

Effect of Constraints

Angle-of-attack limits, actuator bounds, and atmospheric variation introduce perturbations that can be represented as a bounded disturbance,

$$\dot{\rho} = f_0(\rho) + d(t), \qquad |d(t)| \leq d_{\max}. \tag{18}$$

These effects explain transient fluctuations in predicted terminal landing error during atmospheric transition without implying instability.

Interpretation

The above reasoning suggests practical boundedness of LOS landing error under the reduced-order pitch authority representation adopted in this work. No claim is made regarding asymptotic convergence or global stability of the full nonlinear system. The appendix is intended only to provide qualitative insight consistent with the numerical trends presented in the main text.

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